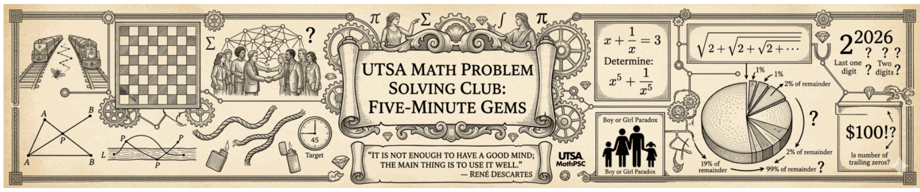

The Handshake Lemma

At a party, several people shake hands. A person cannot shake hands with themselves, and cannot shake the same person's hand twice.

Show that: The number of people who shake hands an odd number of times must be even.

The Mutilated Chessboard

Suppose you have a standard 8×8 chessboard with two diagonally opposite corner squares removed. Is it possible to cover the remaining 62 squares perfectly with 31 dominoes (where each domino covers exactly two adjacent squares)?

The Boy or Girl Paradox

Mr. Smith has two children. At least one of them is a boy.

What is the probability that both children are boys? (Assume boys and girls are equally likely among the general population.)

Values without Solving

Given that $x + \frac{1}{x} = 3$, determine the value of $x^5 + \frac{1}{x^5}$ without solving for x.

The Burning Ropes

You have two ropes. Each rope takes exactly 60 minutes to burn from one end to the other, but they do not burn at a constant rate (i.e., half the rope might burn in 1 minute, and the rest in 59 minutes). **Question:** Using these two ropes and a lighter, how can you measure exactly 45 minutes?

The Fly and the Trains

Two trains are on the same track, 200 miles apart, moving towards each other at a constant speed of 50 mph each. A super-fly starts at the front of one train and flies towards the other at 75 mph. When it reaches the other train, it instantly turns around and flies back. It continues this zig-zag path until the trains collide.

Question: What is the total distance the fly has traveled?

A Symmetric Integral

Evaluate the integrals:

$$\int_{-\pi}^{\pi} (\sin 5x + \sin 3x + \cos x)\, dx, \qquad \int_{-\pi}^{\pi} (\sin^3 x + \sin x)\, dx, \qquad \int_{-\pi}^{\pi} (\sin(x^5) + 1)\, dx.$$

The Infinite Radical

Find the value of

$$\sqrt{2 + \sqrt{2 + \sqrt{2 + \ldots}}}.$$

Bonus: Explain the math behind such “infinitely deep/nested“ square roots.

Trailing Zeros

How many zeros are at the end of the number 100! (100 factorial)?

The River Crossing

A person is at point A and needs to get to point B on the same side of a straight river line L. They must touch the river at some point P to fetch water before going to B. Where should P be located on line L to minimize the total distance $AP + PB$?

Powers of 2

Find the last digit of the number 2^{2026}. How about the last two digits? How about the last three?

Telescoping Series

Evaluate the following sum exactly:

$$\sum_{n=1}^{99} \frac{1}{n^2+n} = \frac{1}{1^2+1} + \frac{1}{2^2+2} + \cdots + \frac{1}{99^2+99}.$$

Bonus: Prove that

$$\sum_{n=1}^{\infty} \frac{1}{n^2} < 2.$$

Can you find the exact value of the sum (series) on the left-hand side?

The largest slice

A huge pie is to be eaten by 100 people. The first person eats 1% of the pie; the second person eats 2% of what is left after the first person; the third person eats 3% of what is left after the first two, and so on, until the 99th person eats 99% of whatever pie is left, and the 100th and last person takes all (100%) of the remaining pie.

Question: Who ate the most pie?